

3.0 Summative Assessment: Network Monitoring and Troubleshooting Scenario

Introduction

Modern business operations depend heavily on reliable network performance to support communication, cloud services, internal applications, and internet access. When network performance becomes unstable, even intermittently, it can negatively affect productivity, business operations, and the user experience. In this scenario, employees are reporting slow internet speeds, dropped connections, and delays when accessing internal resources. As the network administrator, the objective is to use network monitoring technologies to:

1. Identify the root cause of the problem,
2. Restore stable performance, and
3. Implement monitoring solutions that help prevent future issues.

Successfully troubleshooting network problems requires more than simply restarting devices or replacing cables. Administrators must understand how monitoring tools, physical infrastructure, traffic patterns, and communication protocols work together throughout the network. By analyzing network behavior and comparing it against normal operating conditions, administrators can identify performance bottlenecks and develop both immediate and long-term solutions.

Network Monitoring Tools

Several monitoring tools would be used together because each provides a different level of visibility into network performance and device behavior.

Wireshark

Wireshark is a packet analysis tool used to capture and inspect network traffic in real time. It allows administrators to examine communication between devices and identify traffic-related problems. Wireshark would help identify:

- Packet loss
- Excessive retransmissions

- High latency
- DNS failures
- Abnormal traffic patterns
- Dropped connections

This tool is especially useful because it provides detailed visibility into how data is moving across the network and where communication failures may be occurring.

SNMP Monitoring Platforms

Simple Network Management Protocol (SNMP) monitoring systems such as [SolarWinds](#) or [Paessler Router Traffic Grapher \(PRTG\)](#) provide centralized monitoring for routers, switches, servers, firewalls, and wireless access points. These tools allow administrators to monitor overall network health and performance in real time.

Important metrics collected through SNMP include:

- Interface utilization
- CPU and memory usage
- Device uptime
- Error rates
- Packet drops
- Wireless performance statistics

SNMP monitoring is valuable because it combines real-time visibility with long-term historical reporting, allowing administrators to identify both immediate issues and developing trends.

Syslog Monitoring

Syslog servers collect, analyze and store event logs generated by networking devices, servers and applications. Reviewing logs often helps identify problems that may not be immediately visible during live monitoring. Syslog analysis can reveal:

- Interface failures
- Authentication issues
- Routing changes
- Device restarts
- Power-related events
- Wireless connectivity problems

Together, these monitoring systems (Wireshark, SNMP, and Syslog monitoring) provide a complete view of network traffic, device health, and operational stability.

Establishing a Network Baseline and Real-Time Monitoring

A network baseline is a documented record of normal network behavior collected over time. Establishing a baseline allows administrators to compare current conditions against expected network performance and quickly identify abnormal activity.

The baseline process would begin by identifying critical devices, applications, and network segments. Monitoring tools would then collect data during normal business operations across multiple days to establish accurate traffic and performance patterns.

Several important metrics would be recorded, including:

- Bandwidth utilization
- Latency and response times
- Packet loss
- CPU and memory utilization
- Interface statistics
- Wireless signal strength
- Application performance

Once a baseline is established, real-time monitoring would be configured to continuously track network activity and generate alerts when performance exceeds acceptable thresholds.

Several metrics are especially important during real-time monitoring. High bandwidth utilization may indicate congestion, unauthorized downloads, or malicious traffic. Packet loss and latency directly affect application performance and user connectivity. Device CPU and memory usage can reveal overloaded hardware, routing problems, or denial-of-service activity. Wireless signal strength and channel interference are also important because poor wireless coverage often causes dropped connections and inconsistent performance.

By continuously monitoring these conditions, administrators can identify developing issues before they significantly impact users.

Data Collection and Analysis

Network performance data would be collected from routers, switches, firewalls, wireless access points, servers, and high-traffic network segments. Special attention would be given to core switches, internet edge routers, wireless infrastructure, uplink connections, and heavily utilized VLANs because issues in these areas can affect large portions of the network.

Several types of information would be collected and compared together, including:

- SNMP performance statistics
- Packet captures
- Syslog events
- Interface counters
- Wireless performance data
- Application response times

The collected data would then be analyzed to identify patterns and abnormalities. Administrators would look for recurring bandwidth spikes, excessive retransmissions, interface errors, authentication failures, unstable routing behavior, or wireless interference.

For example, if packet captures reveal excessive retransmissions and interface errors on a switch uplink, the issue may involve faulty cabling, duplex mismatches, or overloaded links. Similarly, wireless monitoring data showing overlapping channels and weak signal strength could explain dropped wireless connections and slow internet performance for employees.

The goal of data analysis is not only to identify symptoms but to determine the actual root cause of the performance issues.

Troubleshooting and Resolution

Once the root cause has been identified, troubleshooting would proceed using a structured step-by-step approach designed to restore stable network performance while minimizing disruption to users.

Immediate corrective actions may include:

- Replacing damaged cables
- Correcting duplex mismatches
- Restarting unstable interfaces

- Rebooting malfunctioning devices
- Adjusting wireless channels
- Removing unauthorized devices
- Limiting excessive traffic

For example, if monitoring reveals excessive interface errors, replacing cabling and correcting switch configurations may immediately restore stable communication. If wireless interference is identified, changing wireless channels and repositioning access points may improve coverage and reduce dropped connections.

Long-term solutions would focus on improving network reliability and preventing future performance issues. These improvements may include upgrading bandwidth capacity, replacing outdated hardware, implementing VLAN segmentation, improving wireless coverage, and adding redundant connections.

Physical installation factors would also be reviewed carefully because poor cable management, overheating equipment, unstable power conditions, and improper device placement can contribute to intermittent performance problems.

Important physical installation considerations include:

- Proper cable management
- Environmental controls and cooling
- UPS and surge protection systems
- Physical security for network equipment
- Equipment labeling and documentation
- Redundant power and network links

Proper physical infrastructure supports long-term network stability and simplifies troubleshooting and maintenance.

Historical Data, Documentation, and Communication

Historical monitoring data plays an important role in long-term network management because it allows administrators to identify trends, recurring problems, and future capacity needs. By reviewing historical performance reports, administrators can predict and plan for:

- Bandwidth growth
- Identify recurring failures and devices nearing capacity limits

- Detect performance trends and detect gradual hardware degradation
- Prevent future outages

If historical data, for example, consistently shows increasing bandwidth use during peak business hours, administrators have some decisions to make: to upgrade internet connections or implement traffic prioritization before users experience any significant slowdowns. Historical trend analysis helps organizations move from reactive troubleshooting to proactive network management.

Proper documentation is also essential throughout the troubleshooting process. All findings, monitoring results, packet captures, configuration changes, corrective actions, and recommendations should be recorded clearly. Good documentation improves future troubleshooting, supports operational consistency, reduces downtime, and helps other administrators understand previous issues and solutions.

Communication with stakeholders is equally important. After resolving the issue, administrators should provide management and affected users with a clear explanation of the problem, the steps taken to resolve it, and any long-term improvements being implemented. Stakeholders should also be informed about ongoing monitoring plans that may include:

- Continuous SNMP monitoring
- Automated alerting systems
- Monthly performance reporting
- Wireless site surveys
- Capacity reviews
- Security monitoring

Clear communication helps maintain trust and ensures that both technical staff and business leadership understand how the network is being managed and improved.

Conclusion

Network monitoring technologies play a critical role in identifying and resolving performance problems in modern enterprise networks. Tools such as Wireshark, SNMP monitoring platforms, and Syslog servers provide administrators with visibility into network traffic, device performance, and overall network health.

Establishing a network baseline allows administrators to recognize abnormal behavior quickly, while real-time monitoring helps detect developing issues before they become

major outages. Effective troubleshooting requires careful data collection, structured analysis, and consideration of both logical and physical network infrastructure.

By combining monitoring technologies, proper physical installation practices, historical trend analysis, and detailed documentation, organizations can improve reliability, reduce downtime, and maintain stable network performance for long-term business operations.

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